Class 28 logistic orbit diagram

Activity

Teams 1 and 2: Post photos of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

Questions

1. (Locating a bifurcation)

   The logistic map is given by $f(x) = rx(1-x)$.

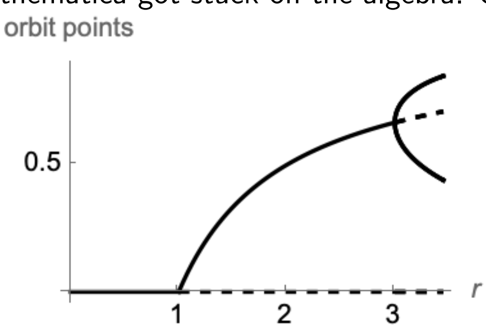
   Sketch this parabola for a few values of r.

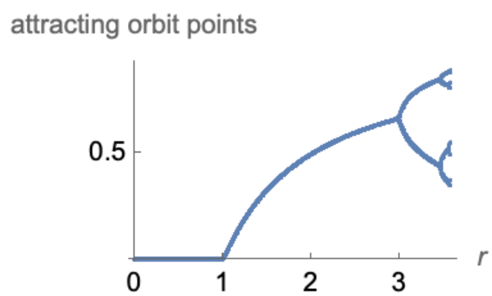
   (a) Find a value of r associated with a tangent bifurcation at $x^* = 0$.

   A nonzero fixed point is born in this bifurcation.

   (b) Confirm that the second fixed point is given by $x^* = \frac{r-1}{r}$. For this fixed point, find a value of r associated with a period-doubling bifurcation.

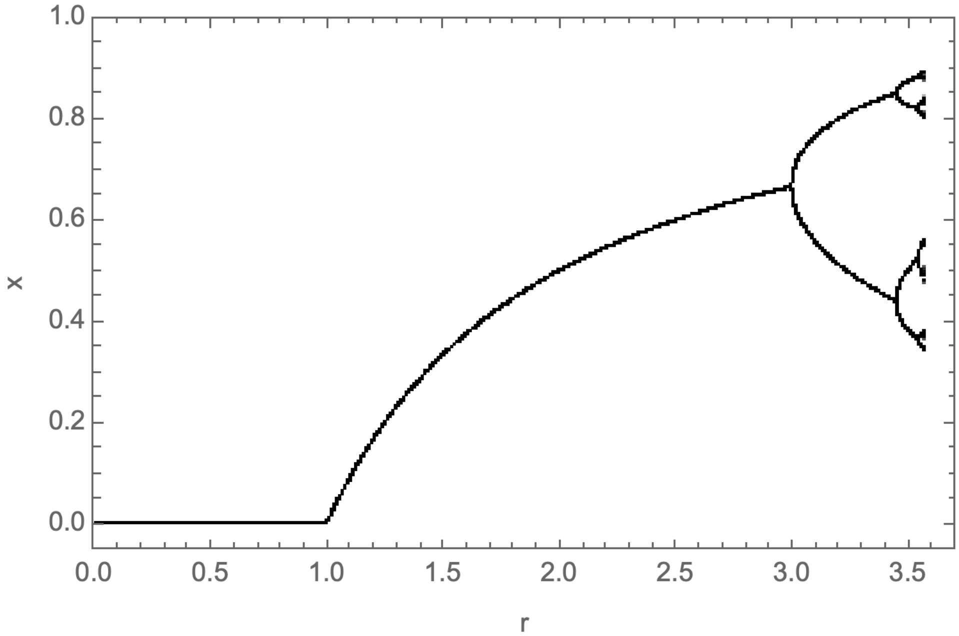
On the left is a bifurcation diagram for the logistic map, $f(x) = rx(1-x)$. Beyond $r = 3.5$ Mathematica got stuck on the algebra. On the right is an orbit diagram for $0 \le r \le 3.56$.





2. (orbit diagram of the logistic map) Below I plot the orbit diagram for the logistic map for $0 \le r \le 3.56$.

   I used a single initial condition at each value of r, then iterating many times, and plotting the last 80 iterates. This means the plotted points you see for a particular value of r are long-term values for a single orbit.



(a) What happens in the orbit diagram at $r = 1$? Relate this to a bifurcation you found above.

(b) What happens at $r = 3$? Relate this to a bifurcation you found above.

(c) If you see one branch, two branches, or four branches, what does that say about the orbit?

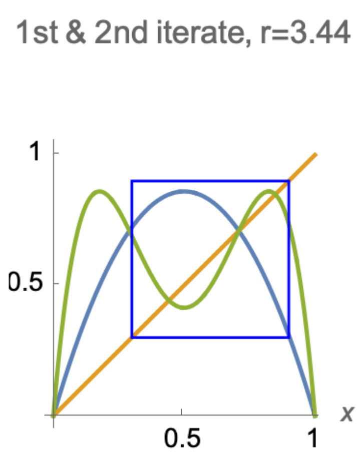
(d) As r increases, what pattern do you see in the number of branches? What kind of sequence does this suggest? *What kind of bifurcation can cause this kind of change in period?*

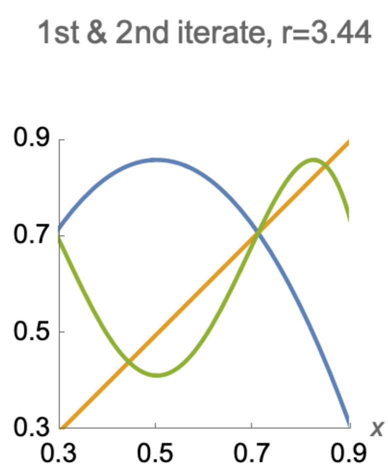
3. (Period doubling in the logistic map)

The logistic map is given by $f(x) = rx(1-x)$.

We will look at the period-doubling bifurcations that cause the period of the stable orbit to double.

(a) The map and its second iterate are given below.

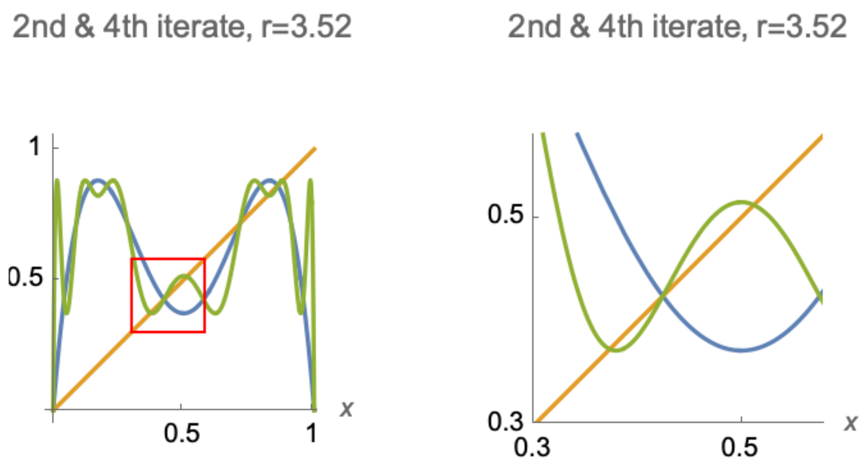




For $r = 3.44$, use the plot above to identify the points involved in the period-2 orbit of the logistic map.

(b) A period-2 orbit of f appears as two fixed points of $f(f(x))$. If this period-2 orbit itself undergoes period-doubling at $r = r_2$, what is the period of the new orbit?

(c) The period-2 orbit has undergone period-doubling by $r = 3.52$ (shown below).



It is hard to distinguish all of the fixed points of $f^4(x)$. In the zoomed-in figure on the right, we can see one period-2 point. How many period-4 points are visible in that zoomed-in region?

They are part of a single period-4 orbit.

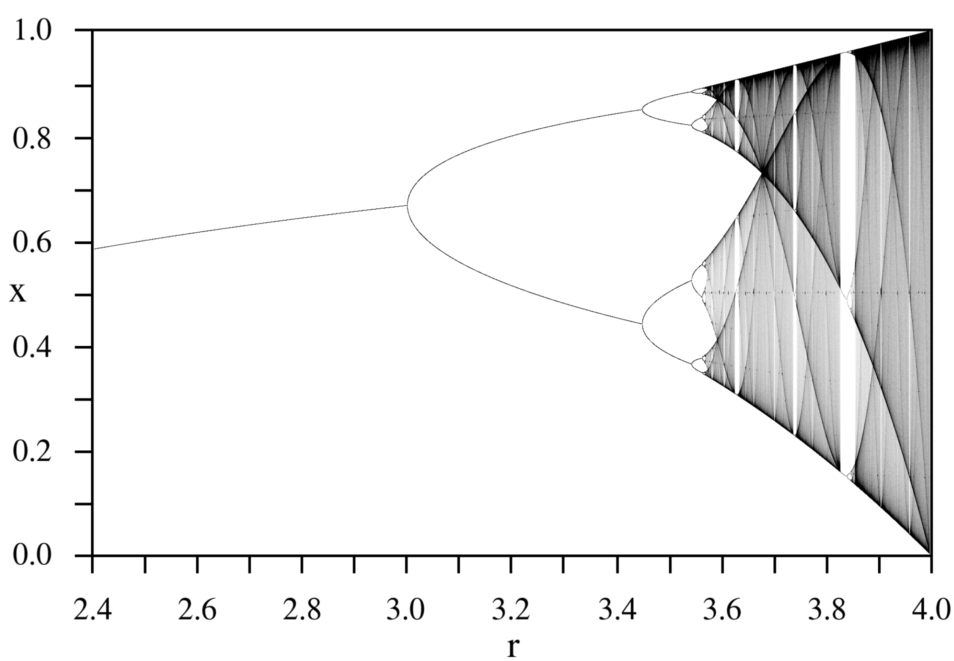
(d) If this happens again, what period would you expect?

As bifurcations continue, this sequence of bifurcations is called a **period-doubling cascade**.

(e) As the cascade continues, how many points will be involved in the attracting orbit?

As repeated doubling creates finer structure what kind of set is the limit of this process?

A more extensive orbit diagram for the logistic map is below. "Fuzzy" areas correspond to values of r where there is a chaotic attractor.



In the logistic map, repeated bifurcations create finer and finer structure in the long-term behavior. To study this kind of limiting set, we will work with a simpler map where the geometry is easier to figure out.

4. (A limiting set as initial conditions escape. Example 4.9, Alligood et al)

   Consider the tent map $x_{n+1} = T_3(x_n)$, defined as

$$x_{n+1} = \begin{cases} 3x_n, & x_n \leq \frac{1}{2}, \\ 3(1 - x_n), & \frac{1}{2} \leq x_n. \end{cases}$$

   This is a slope-3 tent map. Consider it to be defined on the whole real line. Let C be the set of points on the real line whose orbits do not diverge to $-\infty$. Equivalently, C is the set of initial conditions whose iterates under T_3 never become negative.

   We will start to investigate the geometry of this limiting set.

   (a) Sketch the map.

   (b)
   - Identify the fixed points of the map.
   - Identify points that map to zero, so $T_3(x) = 0$.
   - Identify points whose second iterate, $T_3^2(x)$, is zero (these are points that map to points that map to zero).

   These are all points that are part of C.

   (c) Convince your team that initial conditions outside of $[0, 1]$ will all have orbits that eventually diverge. Also convince yourselves that the interval $(1/3, 2/3)$ consists exactly of the points that leave the interval $[0, 1]$ under a single iteration of T_3 (note that $1/3$ and $2/3$ both stay in the interval). Sketch the line segments that are still in consideration for potentially being in C. Call this set C_1.

   (d) What points leave $[0, 1]$ under two iterations of T_3? Again sketch the line segments that are still in consideration for staying in the interval. Call this set C_2. Is this set closed or open? (That is, do the intervals contain their endpoints or not?)

   (e) Can you generalize this to sketch C_3 (three iterations)? What do you think will happen after k iterations?

   (f) What points seem to be in C?